$\mathrm{Hom}_A(A, N) = N$), so the statement follows, since if the direct sum of two complexes is exact, then each of them is exact. To prove that (iv) implies (i), let K be the kernel of the map α, and apply the exact functor $\mathrm{Hom}_A(P, ?)$ to the exact sequence

$$0 \to K \to M \to N \to 0.$$

□

Definition 8.1.2. A module satisfying any of the conditions (i)—(iv) of Theorem 8.1.1 is said to be **projective**.

Problem 8.1.3. A right A-module M is said to be **flat** if the functor $M \otimes_A$ on the category of left A-modules is exact.

(i) Show that any projective module is flat.

(ii) Let A be a commutative ring and let S be any multiplicatively closed subset of A. Then, the localization $S^{-1}A$ is a flat A-module.

(iii) Let $A = \mathbb{C}[x]$, $M = \mathbb{C}[x, x^{-1}]$. Show that M is flat but not projective.

Exercise 8.1.4. Let A be a ring, let M_1, M_2 be left A-modules, let P_1, P_2 be projective left A-modules, and let $f_i : P_i \to M_i$ be homomorphisms. Let M be a left A-module containing M_1 such that $M/M_1 = M_2$. Show that there exists a homomorphism $f : P_1 \oplus P_2 \to M$ such that $f|_{P_1} = f_1$ and the induced homomorphism $P_2 \to M_2$ is f_2.

There is also a notion of an injective module, which is dual to the notion of a projective module. Namely, we have the following theorem.

Let A be an algebra and let I be a left A-module.

Theorem 8.1.5. *The following properties of I are equivalent:*

(i) If $\alpha : N \to M$ is an injective morphism and $\nu : N \to I$ is any morphism, then there exists a morphism $\mu : M \to I$ such that $\mu \circ \alpha = \nu$.

(ii) Any injective morphism $\alpha : I \to M$ splits; i.e., there exists $\mu : M \to I$ such that $\mu \circ \alpha = \mathrm{id}$.